

Electrical Engineering

Comprehensive Guide

Professional Edition 2026

Harz Digital Services

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Chapter 1

Fundamentals of Electrical Engineering

Electrical engineering deals with the study and application of electricity, electronics, and electromagnetism. It is one of the broadest engineering disciplines, powering everything from small consumer devices to national power grids.

Core Concepts

Voltage (V)

Ohm's Law

Ohm's Law

A 12V battery drives 2A through a resistor. $R = V/I = 12/2 = 6$ ohms

Kirchhoff's Laws

KCL: The

AC vs DC Systems

Direct Cu

Chapter 2

Circuit Analysis and Design

Circuit analysis is the process of finding all voltages and currents in a circuit. Engineers use several systematic methods to solve complex circuits.

Series and Parallel Circuits

In a series

Two 10-ohm resistors in parallel: $R = (10 \times 10) / (10 + 10) = 5$ ohms

Nodal Analysis

Nodal analysis involves:
Assign voltage variables to nodes
Write KCL equations for each node

Mesh Analysis

Mesh analysis involves:
Assign current variables to meshes
Write KVL equations for each mesh

Thevenin's Theorem

Any linear circuit containing independent and dependent sources and resistors (or inductors and capacitors) can be replaced by an equivalent circuit consisting of a voltage source in series with a resistor (or inductor or capacitor).

Norton's Theorem

Any linear circuit containing independent and dependent sources and resistors (or inductors and capacitors) can be replaced by an equivalent circuit consisting of a current source in parallel with a resistor (or inductor or capacitor).

Chapter 3

Power Systems and Generation

Electrical power systems generate, transmit, and distribute electricity to consumers.

Power Generation Methods

Thermal:

Three-Phase Power

Three-ph
delivery t

A 480V three-phase motor draws 10A at 0.85 power factor. $P = 1.732 \times 480 \times 10 \times 0.85 = 7,069W$ (7.1kW)

Power Factor Correction

Power fa
inductive

Transformers

Transform
N_primari

Chapter 4

Motors and Drives

Electric motors convert electrical energy into mechanical energy.

Motor Types

DC Moto

Motor Starting Methods

Direct-On

Motor Efficiency

Motor eff

A 5HP motor (3.73kW) draws 4.5kW. Efficiency = $3.73/4.5 = 82.9\%$

VFD Benefits

Variable

Chapter 5

Electrical Safety and Protection

Electrical safety is paramount. Arc flashes, electrocution, and electrical fires cause serious injuries annually.

Protection Devices

Fuses: M

Arc Flash Hazards

Arc flash
arc-rated

Grounding and Bonding

Grounding
difference

Lockout/Tagout (LOTO)

LOTO pr
equipmen
Remove

NEC (National Electrical Code)

NFPA 70

Chapter 6

Renewable Energy Systems

Renewable energy is transforming electrical engineering. Solar, wind, and energy storage are the fastest-growing sectors.

Solar Power Systems

Photovol

A 5kW solar system in Nigeria (5 peak sun hours) generates: $5kW \times 5h \times 0.8 = 20kWh/day$

Wind Power

Wind turbl
speed, C

Energy Storage

Battery to
Sodium-i

Grid Integration

Renewab
response

Net Metering

Net mete

Chapter 7

Electrical Measurements and Instrumentation

Accurate measurement is essential for design, troubleshooting, and quality assurance.

Measurement Instruments

Multimeter

Measurement Accuracy

Accuracy

A multimeter reads 120.5V on a 120.0V standard. Accuracy = $0.5/120.0 \times 100 = 0.42\%$

Transducers and Sensors

Temperature

Data Acquisition (DAQ)

DAQ system
signal frequency

Calibration

Calibration
international

Chapter 8

Power Electronics

Power electronics involves the conversion and control of electrical power using semiconductor devices.

Semiconductor Devices

Diodes: P

Converter Types

Rectifier:

A buck converter converts 24V to 12V at 5A. Input current = $(12 \times 5) / (24 \times 0.9) = 2.78\text{A}$ (90% efficiency)

PWM (Pulse Width Modulation)

PWM con
(for buck

Applications

Variable

Harmonics

THD (Tot

Chapter 9

Control Systems

Control systems regulate the behavior of dynamic systems using feedback.

Open-Loop vs Closed-Loop

Open-loop

Transfer Functions

A transfer

PID Controllers

PID is the

Stability Analysis

A system
Locus, B

For $G(s) = 1/(s^2 + 3s + 2)$, poles are $s = -1$ and $s = -2$ (both negative real), system is stable.

Modern Control

State-space
filter.

Chapter 10

Standards and Codes

Electrical engineering standards ensure safety, reliability, and interoperability.

International Standards

IEC: Glob

Key Standards

IEC 6036

NEMA Ratings

NEMA 1:

IP Ratings

Two-digit
in 1m wa

Professional Licensure

PE licens